

Answers

1. $\frac{1}{z^k}$
2. $\frac{z}{z-1}$, ROC : $|z| > 1$
3. $\frac{z}{z-a}$, ROC : $|z| > |a|$
4. $\frac{1}{a}(1 - e^{-at}) u(t)$
5. $1 - e^{-tR/L}$ for $t \geq 0$
6. $y(n) = 2$ for $n \geq 0$, and $y(n) = 2^{n+1}$ for $n < 0$

One Sided Z-transform

We now focus our attention on one sided Z-transform. If the function $f(t)$ is defined for $t = 0, T, 2T, \dots$, which are discrete values of t , then

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}, \text{ provided that the series converges.}$$

7.10 Z-transform of Some Elementary Functions

1. Z-transform of unity

$$\begin{aligned} Z(1) &= \sum_{n=0}^{\infty} z^{-n} = \sum \left(\frac{1}{z}\right)^n = \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right) = \left(\frac{1}{1 - \frac{1}{z}}\right) \\ &= \frac{z}{z-1} \quad \text{for } |z| > 1 \end{aligned}$$

2. Z-transform of k

$$Z(k) = \sum_{n=0}^{\infty} k z^{-n} = k \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n = k \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right)$$

$$= k \left(\frac{1}{1 - \frac{1}{z}} \right) = \frac{kz}{z-1} \quad \text{for } |z| > 1$$

3. Z-transform of $(-1)^n$

$$Z(-1)^n = \sum_{n=0}^{\infty} (-1)^n z^{-n} = \sum_{n=0}^{\infty} \left(-\frac{1}{z} \right)^n = 1 - \frac{1}{z} + \frac{1}{z^2} - \dots = \frac{z}{z+1}$$

4. Z-transform of a^n

$$Z(a^n) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z} \right)^n = 1 + \frac{a}{z} + \frac{a^2}{z^2} + \dots = \frac{1}{1 - \frac{a}{z}} = \frac{z}{z-a}$$

5. Z-transform of n

$$\begin{aligned} Z(n) &= \sum_{n=0}^{\infty} n \cdot 1 = -z \frac{d}{dz} \{Z(1)\} = -z \frac{d}{dz} \left(\frac{z}{z-1} \right) \\ &= -z \left[\frac{(z-1) \cdot 1 - z}{(z-1)^2} \right] = \frac{z}{(z-1)^2} \end{aligned}$$

6. Z-transform of na^n

$$Z(na^n) = -z \frac{d}{dz} Z(a^n) = -z \frac{d}{dz} \left(\frac{z}{z-a} \right) = \frac{az}{(z-a)^2}$$

7. Z-transform of $\cos n\theta$ and $\sin n\theta$

$$\text{We know that } Z(a^n) = \frac{z}{z-a} \quad \text{if } |z| > |a| \quad (1)$$

$$\text{Let } a = e^{i\theta} \quad \therefore a^n = e^{in\theta}$$

Now, (1) reduces to

$$Z(e^{in\theta}) = \frac{z}{z - e^{i\theta}} = \frac{z}{z - (\cos\theta + i\sin\theta)}$$

$$\begin{aligned} \text{or, } Z(\cos n\theta + i\sin n\theta) &= \frac{z}{(z - \cos\theta) - i\sin\theta} \\ &= \frac{z [(z - \cos\theta) + i\sin\theta]}{[(z - \cos\theta) - i\sin\theta] [(z - \cos\theta) + i\sin\theta]} \end{aligned}$$

$$= \frac{(z^2 - z \cos \theta) + iz \sin \theta}{(z - \cos \theta)^2 + \sin^2 \theta}$$

$$= \frac{(z^2 - z \cos \theta) + iz \sin \theta}{z^2 - 2z \cos \theta + 1}$$

$$\therefore Z(\cos n\theta + i \sin n\theta) = \frac{(z^2 - z \cos \theta)}{z^2 - 2z \cos \theta + 1} + i \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$$

Comparing real and imaginary parts

$$Z(\cos n\theta) = \frac{z^2 - z \cos \theta}{z^2 - 2z \cos \theta + 1} \text{ and } Z(\sin n\theta) = \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$$

Replacing θ by $\frac{\pi}{2}$, we obtain

$$Z\left(\cos \frac{n\pi}{2}\right) = \frac{z^2}{z^2 + 1} \text{ and } Z\left(\sin \frac{n\pi}{2}\right) = \frac{z}{z^2 + 1}$$

Example

Find $Z(r^n \cos n\theta)$ and $Z(r^n \sin n\theta)$.

We have $Z(a^n) = \frac{z}{z - a}$

Let $a = re^{i\theta}$. Then

$$Z(a^n) = Z(r^n e^{in\theta}) = \frac{z}{z - re^{i\theta}}$$

$$\text{or, } Z(r^n \cos n\theta + i r^n \sin n\theta) = \frac{z(z - r \cos \theta) + i(zr \sin \theta)}{z^2 - 2zr \cos \theta + r^2}$$

Now comparing real and imaginary parts

$$Z(r^n \cos n\theta) = \frac{z(z - r \cos \theta)}{z^2 - 2zr \cos \theta + r^2} \text{ and}$$

$$Z(r^n \sin n\theta) = \frac{zr \sin \theta}{z^2 - 2zr \cos \theta + r^2}$$

7.11 Some Important Theorems.

$$1. \quad Z(t^k) = -Tz \frac{d}{dz} Z(t^{k-1})$$

$$\begin{aligned} \text{Here, } Z(t^k) &= \sum_{n=0}^{\infty} (nT)^k z^{-n} = Tz \sum_{n=0}^{\infty} n^k T^{k-1} z^{-(n+1)} \\ &= Tz \sum_{n=0}^{\infty} (nT)^{k-1} n(z)^{-(n+1)} \end{aligned} \quad (1)$$

$$\text{Also, } Z(t^{k-1}) = \sum_{n=0}^{\infty} (nT)^{k-1} z^{-n}$$

Differentiating both sides with respect to z , we get

$$\frac{d}{dz} (Z(t^{k-1})) = \sum_{n=0}^{\infty} (nT)^{k-1} (-n) z^{-n-1} = - \sum_{n=0}^{\infty} (nT)^{k-1} n z^{-(n+1)} \quad (2)$$

Now from (1) and (2)

$$Z(t^k) = -Tz \frac{d}{dz} [Z(t^{k-1})]$$

Special Cases

(1) If $k = 1$, then

$$Z(t) = -Tz \frac{d}{dz} \left(\frac{z}{z-1} \right) = \frac{Tz}{(z-1)^2}$$

(2) If $k = 2$, then

$$Z(t^2) = -Tz \frac{d}{dz} [Z(t)] = -Tz \frac{d}{dz} \left[\frac{Tz}{(z-1)^2} \right] = \frac{T^2 z (z+1)}{(z-1)^3}$$

2. First Shifting Theorem

If $Z\{f(t)\} = F(z)$, then $Z(e^{-at} f(t)) = F(ze^{aT})$

$$\begin{aligned} \text{Here, } Z[e^{-at} f(t)] &= \sum_{n=0}^{\infty} e^{-anT} f(nT) z^{-n} \\ &= \sum_{n=0}^{\infty} f(nT) (ze^{aT})^{-n} \\ &= F(ze^{aT}) \end{aligned}$$

The result can be stated as

If $Z[f(t)] = F(z)$, then $Z[e^{-at} f(t)] = F(z)|_{z \rightarrow ze^{aT}}$.

3. Second Shifting Theorem

If $Z[f(t)] = F(z)$, then $Z[f(t + T)] = z[F(z) - f(0)]$

Here $Z\{f(t + T)\} = \sum_{n=0}^{\infty} f(nT + T) z^{-n} = z \sum_{n=0}^{\infty} f[(n + 1)T] z^{-(n+1)}$

Let $k = n + 1$, then

$$\begin{aligned} Z\{f(t + T)\} &= z \sum_{k=1}^{\infty} f(kT) z^{-k} \\ &= z \left[\sum_{k=0}^{\infty} f(kT) z^{-k} - f(0) \right] \\ &= z [Z f(t) - f(0)] \\ &= z [F(z) - f(0)]. \end{aligned}$$

4. If $Z[f(t)] = F(z)$, then $Z[nf(t)] = -z \frac{d}{dz} F(z)$.

Here $Z[n f(t)] = \sum_{n=0}^{\infty} n f(nT) z^{-n}, \quad (1)$

where $F(z) = Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

Now differentiating with respect to z

$$\frac{d[F(z)]}{dz} = \sum_{n=0}^{\infty} f(nT) (-n) z^{-n-1}$$

or, $\frac{d[F(z)]}{dz} = -z^{-1} \sum_{n=0}^{\infty} f(nT) n z^{-n}$

or, $\frac{d[F(z)]}{dz} = -\frac{1}{z} \sum_{n=0}^{\infty} n f(nT) z^{-n}$

or, $\frac{d[F(z)]}{dz} = -\frac{1}{z} Z[n f(t)]$

or, $Z[nf(t)] = -z \frac{d}{dz} F(z)$.

Example

$$\begin{aligned}
 Z(n^2) &= Z(n.n) \\
 &= -z \frac{d}{dz} [Z(n)] \\
 &= -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right] \\
 &= -z \left[\frac{(z-1)^2 (1) - z \cdot 2(z-1)}{(z-1)^4} \right] \\
 &= -z \left[\frac{z^2 - 2z + 1 - 2z^2 + 2z}{(z-1)^4} \right] \\
 &= \frac{z(z+1)(z-1)}{(z-1)^4} \\
 &= \frac{z(z+1)}{(z-1)^3}
 \end{aligned}$$

5. Initial Value Theorem

If $Z[f(t)] = F(z)$, then $f(0) = \lim_{z \rightarrow \infty} F(z)$

Proof.

$$F(z) = Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n} = f(0) + \frac{f(T)}{z} + \frac{f(2T)}{z^2} + \dots$$

$$\therefore \lim_{z \rightarrow \infty} F(z) = f(0)$$

6. Final Value Theorem

If $Z[f(t)] = F(z)$, then $\lim_{t \rightarrow \infty} f(t) = \lim_{z \rightarrow 0} (z-1) F(z)$

Proof.

$$Z[f(t+T) - f(t)] = \sum_{n=0}^{\infty} [f(n+1)T - f(nT)] z^{-n}$$

$$\text{or, } zF(z) - zF(0) - F(z) = \sum_{n=0}^{\infty} [f(n+1)T - f(nT)] z^{-n}$$

Taking limit to both sides

$$\lim_{z \rightarrow 1} [(z-1)F(z) - zf(0)] = \lim_{z \rightarrow 1} \left[\sum_{n=0}^{\infty} \{f(n+1)T - f(nT)\} z^{-n} \right]$$

$$\text{or, } \lim_{z \rightarrow 1} (z-1)F(z) - f(0) = \sum_{n=0}^{\infty} [f(n+1)T - f(nT)]$$

$$= \lim_{m \rightarrow \infty} \sum_{n=0}^m [f(n+1)T - f(nT)]$$

$$= \lim_{m \rightarrow \infty} [\{f(T) - f(0)\} + \{f(2T) - f(T)\} + \dots + \{f(m+1)T - f(mT)\}]$$

$$= \lim_{m \rightarrow \infty} [f(m+1)T - f(0)]$$

$$= f(\infty) - f(0)$$

$$= \lim_{t \rightarrow \infty} f(t) - f(0)$$

$$\therefore \lim_{z \rightarrow 1} (z-1)F(z) = \lim_{t \rightarrow \infty} f(t)$$

7. If $Z[f(t)] = F(z)$, then $Z[a^n f(t)] = F\left(\frac{z}{a}\right)$

It can be stated as: If $Z[f(t)] = F(z)$, then $Z(a^n f(t)) = F(z)|_{z \rightarrow z/a}$

Proof. $Z[a^n f(t)] = \sum_{n=0}^{\infty} a^n f(nT) z^{-n} = \sum_{n=0}^{\infty} f(nT) \left(\frac{z}{a}\right)^{-n} = F\left(\frac{z}{a}\right)$

Examples

1. Find the Z-transform of

i. $\frac{1}{n}$

ii. $\frac{1}{n(n+1)}$

iii. $\frac{1}{n!}$

iv. n^3

v. $(n+1)a^n$

vi. $u(n-1)$

vii. $a^n \cos bt$ and $a^n \sin bt$

viii. $[3 \cdot 2^n + 4(-1)^n]$

$$\begin{aligned} \text{i. } Z\left(\frac{1}{n}\right) &= \sum_{n=1}^{\infty} \frac{z^{-n}}{n} \\ &= \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \dots \\ &= -\ln\left(1 - \frac{1}{z}\right) = \ln\left(\frac{z}{z-1}\right) \end{aligned}$$

$$\begin{aligned} \text{ii. } Z\left[\frac{1}{n(n+1)}\right] &= Z\left[\frac{1}{n} - \frac{1}{n+1}\right] \\ &= Z\left(\frac{1}{n}\right) - Z\left(\frac{1}{n+1}\right)z \\ &= \log\left(\frac{z}{z-1}\right) - \left\{1 + \frac{1}{2z} + \frac{1}{3z^2} + \dots\right\} \\ &= \log\left(\frac{z}{z-1}\right) - z\left(\frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \dots\right) \\ &= \log\left(\frac{z}{z-1}\right) - z\left\{-\log\left(1 - \frac{1}{z}\right)\right\} \\ &= \log\left(\frac{z}{z-1}\right) + z \log\left(\frac{z-1}{z}\right) \\ &= (z-1) \log\left(\frac{z-1}{z}\right) \end{aligned}$$

$$\begin{aligned} \text{iii. } Z\left(\frac{1}{n!}\right) &= \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} \\ &= 1 + \left(\frac{1}{z}\right) + \frac{1}{2!}\left(\frac{1}{z}\right)^2 + \dots \\ &= e^{1/z} \end{aligned}$$

$$\begin{aligned} \text{iv. } Z(n^3) &= Z(n \cdot n^2) \\ &= -z \left[\frac{d}{dz} (Z(n^2)) \right] \end{aligned}$$

$$\begin{aligned}
 &= -z \frac{d}{dz} \left[\frac{z^2 + z}{(z-1)^3} \right] \quad \left[\because Z(n^2) = -z \frac{d}{dz} Z(n) = -z \frac{d}{dz} \left(\frac{z}{(z-1)^2} \right) \right] \\
 &= -z \left[\frac{(z-1)^3 (2z+1) - (z^2+z) 3(z-1)^2}{(z-1)^6} \right] \\
 &= \frac{z^3 + 4z^2 + z}{(z-1)^4}
 \end{aligned}$$

v. $Z\{(n+1)a^n\} = Z(na^n) + Z(a^n)$

$$\begin{aligned}
 &= -z \frac{d}{dz} [Z(a^n)] + \frac{z}{z-a} \\
 &= -z \left[\frac{d}{dz} \left(\frac{z}{z-a} \right) \right] + \left(\frac{z}{z-a} \right) \\
 &= -z \left[\frac{(z-a) - z}{(z-a)^2} \right] + \frac{z}{z-a} \\
 &= \frac{az}{(z-a)^2} + \frac{z}{z-a} \\
 &= \frac{az + z^2 - az}{(z-a)^2} = \frac{z^2}{(z-a)^2}
 \end{aligned}$$

vi. $Z[u(n-1)] = \sum_{n=1}^{\infty} 1 \cdot z^{-n} = \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$

$$= \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right)$$

$$= \frac{1}{z} \left(\frac{1}{1 - \frac{1}{z}} \right) = \frac{1}{z-1}$$

vii. Here, $Z(a^n \cdot e^{ibt}) = [Z(e^{ibt})]_{z \rightarrow z/a}$

or, $Z(a^n \cos bt + ia^n \sin bt) = \left(\frac{z}{z - e^{ibt}} \right)_{z \rightarrow z/a}$

$$\text{or, } Z[a^n (\cos bt + i \sin bt)] = \frac{\frac{z}{a}}{\frac{z}{a} - e^{ibT}}$$

$$\text{or, } Z[a^n (\cos bt + i \sin bt)] = \frac{z}{z - a(\cos bT + i \sin bT)}$$

$$\text{or, } Z[a^n (\cos bt + i \sin bt)] = \frac{z}{(z - a \cos bT) - i(a \sin bT)}$$

$$\begin{aligned} \text{or, } Z[a^n (\cos bt + i \sin bt)] &= \frac{z[(z - a \cos bT) + i(a \sin bT)]}{[(z - a \cos bT) - i(a \sin bT)][(z - a \cos bT) + i(a \sin bT)]} \\ &= \frac{z(z - a \cos bT) + iz(a \sin bT)}{(z - a \cos bT)^2 + (a \sin bT)^2} \end{aligned}$$

$$\begin{aligned} \text{or, } Z(a^n \cos bt) + iZ(a^n \sin bt) &= \frac{z(z - a \cos bT)}{(z - a \cos bT)^2 + (a \sin bT)^2} + \frac{iz a \sin bT}{(z - a \cos bT)^2 + (a \sin bT)^2} \\ &= \frac{z(z - a \cos bT)}{z^2 - 2az \cos bT + a^2} + \frac{iaz \sin bT}{z^2 - 2az \cos bT + a^2} \end{aligned}$$

Comparing real and imaginary parts, we have

$$Z(a^n \cos bt) = \frac{z(z - a \cos bT)}{z^2 - 2az \cos bT + a^2} \quad \text{and}$$

$$Z(a^n \sin bt) = \frac{az \sin bT}{z^2 - 2az \cos bT + a^2}$$

$$\text{viii. } Z[3 \cdot 2^n + 4(-1)^n] = 3Z(2^n) + 4Z(-1)^n$$

$$= 3 \frac{z}{z - 2} + 4 \frac{z}{z + 1}$$

$$= \frac{3z}{z - 2} + \frac{4z}{z + 1}$$

Uses of First Shifting Theorem

2. Find the Z-transform of

(i) e^{-at}

(iii) $e^{-at} \cos bt$

(v) $t^3 e^{-2t}$

(ii) $t e^{-at}$

(iv) $e^{-at} \sin bt$

(vi) $\cosh at \sin bt$

i. $Z(e^{-at}) = Z(e^{-at} \cdot 1)$

$$= [Z(1)]_{z \rightarrow ze^{aT}}$$

$$= \left[\frac{z}{z-1} \right]_{z \rightarrow ze^{aT}}$$

$$= \frac{ze^{aT}}{ze^{aT} - 1}$$

$$\therefore Z(e^{-at}) = \frac{ze^{aT}}{ze^{aT} - 1} = \frac{z}{z - e^{-aT}}$$

ii. $Z(te^{-at}) = [Z(t)]_{z \rightarrow ze^{aT}}$

$$= \left[\frac{Tz}{(z-1)^2} \right]_{z \rightarrow ze^{aT}}$$

$$= \frac{Tz e^{aT}}{(ze^{aT} - 1)^2}$$

$$= \frac{Tz e^{-aT}}{(z - e^{-aT})^2}$$

iii. and iv.

We have $Z(e^{-iat}) = \frac{z}{z - e^{-iaT}}$

or, $Z(\cos at - i \sin at) = \frac{z}{z - e^{-iaT}} \times \frac{z - e^{iaT}}{z - e^{iaT}}$

$$\text{or, } Z(\cos at - i \sin at) = \frac{z(z - \cos aT - i \sin aT)}{z^2 - 2z \cos aT + 1}$$

Comparing real and imaginary parts we get

$$Z(\cos at) = \frac{z(z - \cos aT)}{z^2 - 2z \cos aT + 1}$$

$$\text{and } Z(\sin at) = \frac{z \sin aT}{z^2 - 2z \cos aT + 1}$$

$$\text{Now, } Z(e^{-at} \cos bt) = [Z(\cos bt)]_{z \rightarrow ze^{aT}}$$

$$= \left[\frac{z(z - \cos bT)}{z^2 - 2z \cos bT + 1} \right]_{z \rightarrow ze^{aT}}$$

$$= \frac{ze^{aT} (ze^{aT} - \cos bT)}{z^2 e^{2aT} - 2ze^{aT} \cos bT + 1}$$

$$\text{and } Z(e^{-at} \sin bt) = \left[\frac{z \sin bT}{z^2 - 2z \cos bT + 1} \right]_{z \rightarrow ze^{aT}}$$

$$= \frac{ze^{aT} \sin bT}{z^2 e^{2aT} - 2ze^{aT} \cos bT + 1}$$

$$\text{v. } Z(t^3 e^{-2t}) = [Z(t^3)]_{z \rightarrow ze^{2T}}$$

$$= T^3 \left[\frac{z^3 + 4z^2 + z}{(z-1)^4} \right]_{z \rightarrow ze^{2T}}$$

$$= \frac{T^3 [z^3 e^{6T} + 4z^2 e^{4T} + ze^{2T}]}{(ze^{2T} - 1)^4}$$

$$\text{vi. } Z(\cosh at \sin bt) = Z \left[\frac{1}{2} \times (e^{at} + e^{-at}) \sin bt \right]$$

$$= \frac{1}{2} [Z(e^{at} \sin bt) + Z(e^{-at} \sin bt)]$$

$$= \frac{1}{2} [Z(\sin bt)]_{z \rightarrow ze^{-aT}} + \frac{1}{2} [Z(\sin bt)]_{z \rightarrow ze^{aT}}$$

$$\begin{aligned}
 &= \frac{1}{2} \left[\frac{z \sinh T}{z^2 - 2z \cosh T + 1} \right]_{z=ze^{aT}} + \frac{1}{2} \left[\frac{z \sinh T}{z^2 - 2z \cosh T + 1} \right]_{z=ze^{-aT}} \\
 &= \frac{1}{2} \left[\left\{ \frac{ze^{aT} \sinh T}{(ze^{aT})^2 - 2(ze^{aT}) \cosh T + 1} \right\} + \left\{ \frac{ze^{-aT} \sinh T}{(ze^{-aT})^2 - 2(ze^{-aT}) \cosh T + 1} \right\} \right] \\
 &= \frac{1}{2} \left[\frac{ze^{aT} \sinh T}{z^2 e^{2aT} - 2ze^{aT} \cosh T + 1} + \frac{ze^{-aT} \sinh T}{z^2 e^{-2aT} - 2ze^{-aT} \cosh T + 1} \right]
 \end{aligned}$$

8. Shifting Theorem

If $f(n) = 0$ for $n < 0$ such that $Z[f(n)] = F(z)$ then
 $Z[f(n - kT)] = z^{-k} F(z)$

Proof.

By definition

$$F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} \quad \text{and}$$

$$\begin{aligned}
 Z[f(n - kT)] &= \sum_{n=0}^{\infty} f(nT - kT) z^{-n} \\
 &= \sum_{n=0}^{\infty} f[(n - k)T] z^{-n}
 \end{aligned}$$

Let $n - k = N$

If $n = 0$, $N = -k$

and when $n = \infty$, $N = \infty$

$$\begin{aligned}
 \therefore Z[f(n - kT)] &= \sum_{N=-k}^{\infty} f(NT) z^{-(N+k)} \\
 &= \sum_{N=-k}^{\infty} f(NT) z^{-N} z^{-k} \\
 &= z^{-k} \sum_{N=-k}^{\infty} f(NT) z^{-N} \\
 &= z^{-k} \left[\sum_{N=-k}^{\infty} f(NT) z^{-N} \right] \quad [f(NT) = 0 \text{ for } N < 0] \\
 &= z^{-k} F(z)
 \end{aligned}$$

$$\therefore Z[f(n - kT)] = z^{-k} F(z)$$

List of Formulae

	$f(n)$	$Z[f(n)]$
1.	1 or u_n	$\frac{z}{z-1}$
2.	$(-1)^n$	$\frac{z}{z+1}$
3.	$\delta(n)$	1
4.	$a^n u_n$ or a^n	$\frac{z}{z-a}$
5.	n	$\frac{z}{(z-1)^2}$
6.	n^2	$\frac{z^2+z}{(z-1)^3}$
7.	na^n	$\frac{az}{(z-a)^2}$
8.	$\cos n\theta$ or $u(n) \cos n\theta$	$\frac{z^2 - z \cos \theta}{z^2 - 2z \cos \theta + 1}$
9.	$\sin n\theta$ or $u(n) \sin n\theta$	$\frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$
10.	$r^n \cos n\theta$	$\frac{z(z - r \cos \theta)}{z^2 - 2zr \cos \theta + r^2}$
11.	$r^n \sin n\theta$	$\frac{rz \sin \theta}{z^2 - 2zr \cos \theta + r^2}$
12.	$\frac{1}{n}$	$\ln \left(\frac{z}{z-1} \right)$
13.	$\cos \frac{n\pi}{2}$ and $\sin \frac{n\pi}{2}$	$\frac{z^2}{z^2+1}$ and $\frac{z}{z^2+1}$
14.	$a^n \cos \frac{n\pi}{2}$ and $a^n \sin \frac{n\pi}{2}$	$\frac{z^2}{z^2+a^2}$ and $\frac{az}{z^2+a^2}$
15.	$\frac{1}{n!}$	$e^{1/z}$

	$f(n)$	$Z[f(n)]$
16.	t	$\frac{Tz}{(z-1)^2}$
17.	t^2	$\frac{T^2 z (z+1)}{(z-1)^3}$
18.	t^3	$\frac{T^3 (z^3 + 4z^2 + z)}{(z-1)^4}$
19.	e^{-at}	$\frac{z}{z - e^{-aT}} \quad z > e^{-aT}$
20.	e^{at}	$\frac{z}{z - e^{aT}} \quad z > e^{aT}$
21.	Cosat	$\frac{z(z - \text{cosa}T)}{z^2 - 2z\text{cosa}T + 1}$
22.	Sinat	$\frac{zsinaT}{z^2 - 2z\text{cosa}T + 1}$

Some Further Examples

1. Find the Z-transform of

(i) e^n

(ii) $n3^n$

(iii) $t^2 e^{-bt}$

(iv) na^{n-1}

(v) $[3(2)^n - 4(3)^n] U_n$

i. $Z(e^n) = \sum_{n=0}^{\infty} e^n z^{-n}$

$$= \sum_{n=0}^{\infty} \left(\frac{e}{z}\right)^n$$

$$= 1 + \frac{e}{z} + \left(\frac{e}{z}\right)^2 + \left(\frac{e}{z}\right)^3 + \dots$$

$$= \frac{1}{1 - \frac{e}{z}}$$

$$= \frac{z}{z - e}$$

$$\therefore Z(e^n) = \frac{z}{z-e}$$

$$\begin{aligned} \text{ii. } Z(n3^n) &= \sum_{n=0}^{\infty} n3^n z^{-n} \\ &= \sum_{n=0}^{\infty} n \left(\frac{3}{z}\right)^n \\ &= \left(\frac{3}{z}\right) + 2\left(\frac{3}{z}\right)^2 + 3\left(\frac{3}{z}\right)^3 + \dots \\ &= \frac{3}{z} \left\{ 1 + 2\left(\frac{3}{z}\right) + 3\left(\frac{3}{z}\right)^2 + \dots \right\} \\ &= \frac{3}{z} \left[\frac{1}{\left(1 - \frac{3}{z}\right)^2} \right] \\ &= \frac{3z}{(z-3)^2} \end{aligned}$$

$$\begin{aligned} \text{iii. } Z(t^2 e^{-bt}) &= \frac{T^2 z (z+1)}{(z-1)^3} \Bigg|_{z \rightarrow ze^{bT}} \quad (\text{by First Shifting Theorem}) \\ &= \frac{T^2 ze^{bT} (ze^{bT} + 1)}{(ze^{bT} - 1)^3} \end{aligned}$$

$$\begin{aligned} \text{iv. } Z(na^{n-1}) &= \frac{1}{a} Z(na^n) \\ &= \frac{1}{a} Z\{n\} \Bigg|_{z \rightarrow \frac{z}{a}} \\ &= \frac{1}{a} \frac{z}{(z-1)^2} \Bigg|_{z \rightarrow \frac{z}{a}} \\ &= \frac{1}{a} \frac{\frac{z}{a}}{\left(\frac{z}{a} - 1\right)^2} \\ &= \frac{z}{(z-a)^2} \end{aligned}$$

$$\begin{aligned}
 \text{v. } Z[3(2)^n - 4(3)^n]u_n &= 3Z\{u(n) 2^n\} - 4Z\{u(n) 3^n\} \\
 &= 3 \frac{z}{z-1} \Big|_{z \rightarrow \frac{z}{2}} - 4 \frac{z}{z-1} \Big|_{z \rightarrow \frac{z}{3}} \\
 &= 3 \frac{\frac{z}{2}}{\frac{z}{2}-2} - 4 \frac{\frac{z}{3}}{\frac{z}{3}-3} \\
 &= \frac{3z}{z-2} - \frac{4z}{z-3} \\
 &= \frac{3z^2 - 9z - 4z^2 + 8z}{(z-2)(z-3)} \\
 &= \frac{-z^2 - z}{(z-2)(z-3)}
 \end{aligned}$$

7.12 Convolution

Let $f(t)$ and $g(t)$ be two functions of t , then the convolution of $f(t)$ and $g(t)$, denoted by $f * g$ is defined as follows.

$$f(t) * g(t) = \sum_{k=0}^{\infty} f(kT) g(n-k)T]$$

Theorem.

If $F(z)$ and $G(z)$ be the Z-transforms of $f(t)$ and $g(t)$, then

$$Z[f(t) * g(t)] = F(z) G(z)$$

Proof.

$$\text{We have, } F(z) = Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$G(z) = Z[g(t)] = \sum_{m=0}^{\infty} g(mT) z^{-m}$$

$$\begin{aligned}
 \text{Now, } F(z) G(z) &= \left[\sum_{n=0}^{\infty} f(nT) z^{-n} \right] \left[\sum_{m=0}^{\infty} g(mT) z^{-m} \right] \\
 &= [f(0) + f(T) z^{-1} + f(2T) z^{-2} + \dots] \left[\sum_{m=0}^{\infty} g(mT) z^{-m} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \sum_{m=0}^{\infty} f(0) g(mT) z^{-m} + \sum_{m=0}^{\infty} f(T) g(mT) z^{-m-1} \\
 &\quad + \sum_{m=0}^{\infty} f(2T) g(mT) z^{-m-2} + \dots \\
 &= \sum_{m=0}^{\infty} f(0) g(mT) z^{-m} + \sum_{m=1}^{\infty} f(T) g[(m-1)T] z^{-m} \\
 &\quad + \sum_{m=2}^{\infty} f(2T) g[(m-2)T] z^{-m} + \dots
 \end{aligned}$$

Since $g(-T) = g(-2T) = \dots = 0$, then

$$\begin{aligned}
 F(z) G(z) &= \sum_{m=0}^{\infty} f(0) g(mT) z^{-m} + \sum_{m=0}^{\infty} f(T) g[(m-1)T] z^{-m} + \dots \\
 &= \sum_{m=0}^{\infty} [f(0) g(mT) + f(T) g(m-1)T + \dots] z^{-m} \\
 &= \sum_{m=0}^{\infty} \left[\sum_{k=0}^{\infty} f(kT) g(m-k)T \right] z^{-m}
 \end{aligned}$$

$$\therefore F(z) G(z) = \sum_{m=0}^{\infty} (f * g) z^{-m} = Z(f * g)$$

$$\therefore Z[f * g] = F(z) G(z)$$

7.13 Inverse of Z-transform

If $F(z)$ is the Z-transform of $f(n)$, then the inverse of $F(z)$ is $f(n)$. The inverse of Z-transform is denoted by Z^{-1} . Thus,

$$Z^{-1} [F(z)] = f(n)$$

For example,

$$Z(n) = \frac{z}{(z-1)^2}$$

$$\therefore Z^{-1} \left[\frac{z}{(z-1)^2} \right] = n.$$

Methods of Finding Inverse of Z-transform

1. Long Division Method

Here, the inverse of Z-transform is obtained by actual division in which the given function is expressed as a rational complex

function into infinite power of z^{-1} . Then compare the coefficients with $\sum_{n=0}^{\infty} f(n) z^{-n}$. Finally find the formula for $f(n)$.

Examples

1. Find the inverse of Z-transform of $\frac{1}{z-2}$.

We have $F(z) = \frac{1}{z-2}$

or,
$$\begin{array}{r} z-2 \overline{) \begin{array}{l} 1 \\ 1 \end{array} } \begin{array}{l} z^{-1} + 2z^{-2} + 4z^{-3} \\ - \quad + \\ \hline 2z^{-1} \\ 2z^{-1} \quad - 4z^{-2} \\ - \quad + \\ \hline 4z^{-2} \\ 4z^{-2} \quad - 8z^{-3} \\ - \quad + \\ \hline 8z^{-3} \quad \dots\dots \\ \vdots \\ \text{and so on.} \end{array}$$

Thus, we have

$$\begin{aligned} F(z) &= z^{-1} + 2z^{-2} + 4z^{-3} + \dots\dots \\ &= z^{-1} + 2z^{-2} + 2^2 z^{-3} + \dots\dots \end{aligned}$$

$$\Rightarrow F(z) = \sum_{n=0}^{\infty} 2^{n-1} z^{-n}$$

$$\therefore Z^{-1} F(z) = 2^{n-1}$$

2. Find the $Z^{-1} \left[\frac{z}{(z-2)(z-3)} \right]$

$$\text{We have } \frac{z}{(z-2)(z-3)} = \frac{\frac{z}{z^2}}{\frac{z^2}{z^2} - \frac{5z}{z^2} + \frac{6}{z^2}}$$

$$= \frac{z^{-1}}{1 - 5z^{-1} + 6z^{-2}}$$

Now by actual division

$$\begin{array}{r}
 1 - 5z^{-1} + 6z^{-2} \overline{) z^{-1} + 2z^{-2} + 4z^{-3}} \\
 \underline{z^{-1} - 5z^{-2} + 6z^{-3}} \phantom{+ 4z^{-3}} \\
 5z^{-2} - 6z^{-3} \phantom{+ 4z^{-3}} \\
 \underline{5z^{-2} - 25z^{-3} + 30z^{-4}} \phantom{+ 4z^{-3}} \\
 19z^{-3} - 30z^{-4} \phantom{+ 4z^{-3}} \\
 \underline{19z^{-3} - 95z^{-4} + 114z^{-5}} \phantom{+ 4z^{-3}} \\
 -125z^{-4} - 114z^{-5} \phantom{+ 4z^{-3}} \\
 \phantom{-125z^{-4} - 114z^{-5}} \text{and so on.}
 \end{array}$$

$$\therefore F(z) = z^{-1} + 5z^{-2} + 19z^{-3} + \dots$$

Now comparing with

$$\sum_{n=0}^{\infty} f(n) z^{-n} = f(0) + z^{-1} f(1) + z^{-2} f(2) + z^{-3} f(3) + \dots$$

$$f(0) = 0 \quad f(1) = 1 \quad f(2) = 5 \quad f(3) = 19$$

$$\therefore f(n) = 3^n - 2^n$$

$$\Rightarrow Z^{-1} F(z) = 3^n - 2^n \text{ where } F(z) = \frac{z}{(z-2)(z-3)}$$

2. Partial Fraction Method

This is the method of obtaining Z-transform by resolving $F(z)$ into partial fractions. If $F(z)$ is of the form

i. $F(z) = \frac{H(z)}{(az+b)(cz+d)}$, then it is expressed as

$$\frac{H(z)}{(az+b)(cz+d)} = \frac{Az}{az+b} + \frac{Bz}{cz+d}$$

ii. $F(z) = \frac{H(z)}{(az+b)(cz^2+d)}$, then it is expressed as

$$\frac{H(z)}{(az+b)(cz^2+d)} = \frac{Az}{az+b} + \frac{Bz^2+Cz}{cz^2+d}$$

iii. $F(z) = \frac{H(z)}{(az+b)(cz+d)^2}$ then it is expressed as

$$\frac{H(z)}{(az+b)(cz+d)^2} = \frac{Az}{az+b} + \frac{Bz}{cz+d} + \frac{Cz}{(cz+d)^2}$$

Examples

1. Find the inverse Z-transform of $F(z) = \frac{z}{z^2 + 9z + 20}$.

$$\text{Here, } F(z) = \frac{z}{z^2 + 9z + 20} = \frac{z}{(z+4)(z+5)}$$

$$\Rightarrow \text{Let } F(z) = \frac{Az}{z+4} + \frac{Bz}{z+5}$$

$$\text{or, } \frac{z}{(z+4)(z+5)} = \frac{Az(z+5) + Bz(z+4)}{(z+4)(z+5)}$$

$$\text{or, } z = Az(z+5) + Bz(z+4)$$

$$\text{On solving } A = 1, B = -1$$

$$F(z) = \frac{z}{z+4} - \frac{z}{z+5}$$

$$\text{or, } F(z) = \frac{z}{z - (-4)} - \frac{z}{z - (-5)}$$

$$\text{Now, } Z^{-1}[F(z)] = Z^{-1}\left[\frac{z}{z - (-4)}\right] - Z^{-1}\left[\frac{z}{z - (-5)}\right] = (-4)^n - (-5)^n$$

$$\therefore f(n) = (-4)^n - (-5)^n$$



$$2. \text{ Find } Z^{-1}\left(\frac{z}{(z+2)(z-1)^2}\right).$$

$$\text{Here, } F(z) = \frac{z}{(z+2)(z-1)^2}$$

$$\text{Let } \frac{z}{(z+2)(z-1)^2} = \frac{Az}{z+2} + \frac{Bz}{z-1} + \frac{Cz}{(z-1)^2}$$

$$\text{or, } \frac{z}{(z+2)(z-1)^2} = \frac{Az(z-1)^2 + Bz(z-1)(z+2) + Cz(z+2)}{(z+2)(z-1)^2}$$

$$\text{or, } z = Az(z-1)^2 + Bz(z-1)(z+2) + Cz(z+2)$$

$$\text{If } z = -2, A = \frac{1}{9}; \text{ and if } z = 1, C = \frac{1}{3}$$

Now comparing the coefficients of z^3

$$A + B = 0 \Rightarrow B = -\frac{1}{9}$$

$$\therefore F(z) = \frac{z}{9(z+2)} - \frac{z}{9(z-1)} + \frac{z}{3(z-1)^2}$$

$$\begin{aligned} \Rightarrow Z^{-1}[F(z)] &= \frac{1}{9} Z^{-1} \left[\frac{z}{z-(-2)} \right] - \frac{1}{9} Z^{-1} \left[\frac{z}{z-1} \right] + \frac{1}{3} Z^{-1} \left[\frac{z}{(z-1)^2} \right] \\ &= \frac{1}{9} (-2)^n - \frac{1}{9} + \frac{1}{3} n \end{aligned}$$

$$3. \text{ Find } Z^{-1} \left[\frac{2z}{(z-1)(z^2+1)} \right].$$

$$\text{Let } \frac{2z}{(z-1)(z^2+1)} = \frac{Az}{z-1} + \frac{Bz^2+Cz}{z^2+1}$$

$$\text{or, } 2z = Az(z^2+1) + (Bz^2+Cz)(z-1)$$

$$\text{or, } 2z = (A+B)z^3 + (C-B)z^2 + (A-C)z$$

$$\text{If } z = 1, A = 1.$$

Comparing coefficients of z^3

$$A + B = 0 \Rightarrow A = -B$$

$$\therefore B = -1$$

and by comparing coefficients of z^2

$$C - B = 0 \Rightarrow C = B. \text{ But } B = -1$$

$$\therefore C = -1$$

$$\therefore Z^{-1} \left[\frac{2z}{(z-1)(z^2+1)} \right] = Z^{-1} \left[\frac{z}{z-1} \right] - Z^{-1} \left[\frac{z^2+z}{z^2+1} \right]$$

$$= 1 - \left[\cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

4. Find $Z^{-1} \left[\frac{z^2+z}{(z-2)(z^2+1)} \right]$.

Let $\frac{z^2+z}{(z-2)(z^2+1)} = \frac{Az}{(z-2)} + \frac{Bz^2+Cz}{(z^2+1)}$

or, $z^2+z = Az(z^2+1) + (Bz^2+Cz)(z-2)$

Now putting $z=2$, we get

$$6 = 10A \Rightarrow A = \frac{6}{10} = \frac{3}{5}$$

We have

$$z^2+z = (A+B)z^3 + (-2B+C)z^2 + (A-2C)z$$

Comparing the coefficients of z^3

$$A+B=0$$

$$\Rightarrow B = -A = -\frac{3}{5}$$

and comparing the coefficients of z^2

$$1 = -2B + C$$

$$\text{or, } 1 = -2\left(-\frac{3}{5}\right) + C$$

$$\text{or, } C = 1 - \frac{6}{5} = -\frac{1}{5}$$

$$\begin{aligned} \therefore \frac{z^2+z}{(z-2)(z^2+1)} &= \frac{3}{5} \frac{z}{(z-2)} + \frac{-\frac{3}{5}z^2 - \frac{1}{5}z}{z^2+1} \\ &= \frac{3}{5} \frac{z}{z-2} - \frac{1}{5} \left(\frac{3z^2+z}{z^2+1} \right) \\ &= \frac{3}{5} \frac{z}{z-2} - \frac{3}{5} \frac{z^2}{z^2+1} - \frac{1}{5} \frac{z}{z^2+1} \\ &= \frac{3}{5} (2)^0 - \frac{3}{5} \cos \frac{n\pi}{2} - \frac{1}{5} \sin \frac{n\pi}{2} \end{aligned}$$

3. Residue Method

The inverse Z-transform can also be obtained by using the calculus of residues. If $F(z)$ is the Z-transform of $f(n)$, then

$$f(n) = \frac{1}{2\pi i} \int_c F(z) z^{n-1} dz$$

where c is the closed contour which contains all the poles. By Cauchy Residue Theorem,

$$\int_c F(z) z^{n-1} dz = 2\pi i \text{ times sum of the residues, so}$$

$$f(n) = \text{sum of residues of the integrand at the poles.}$$

Examples

1. Find the inverse of Z-transform of $\frac{z}{z^2 - 3z + 2}$.

Let $F(z) = \frac{z}{z^2 - 3z + 2}$. Then

$$\begin{aligned} f(n) &= \frac{1}{2\pi i} \int_c z^{n-1} F(z) dz \\ &= \frac{1}{2\pi i} \int_c z^{n-1} \left(\frac{z}{z^2 - 3z + 2} \right) dz \\ &= \frac{1}{2\pi i} \int_c \left(\frac{z^n}{z^2 - 3z + 2} \right) dz \\ &= \frac{1}{2\pi i} \int_c \frac{z^n}{(z-1)(z-2)} dz \end{aligned}$$

Clearly, the poles are $z = 1$ and $z = 2$

$$\text{Now residue } (R_1)_{z=1} = \lim_{z \rightarrow 1} (z-1) \left[\frac{z^n}{(z-1)(z-2)} \right] = \frac{1}{-1} = -1$$

$$\text{and residue } (R_2)_{z=2} = \lim_{z \rightarrow 2} (z-2) \left[\frac{z^n}{(z-1)(z-2)} \right] = \frac{2^n}{1} = 2^n$$

$$\therefore f(n) = \text{sum of residues} = -1 + 2^n = 2^n - 1$$

2. Find $Z^{-1} \left(\frac{z}{z^2 + 2z + 4} \right)$.

Let $F(z) = \frac{z}{z^2 + 2z + 4}$, then

$$f(n) = \frac{1}{2\pi i} \int_C z^{n-1} F(z) dz$$

$$= \frac{1}{2\pi i} \int_C z^{n-1} \left(\frac{z}{z^2 + 2z + 4} \right) dz$$

$$= \frac{1}{2\pi i} \int_C \left(\frac{z^n}{z^2 + 2z + 4} \right) dz = \text{sum of residues}$$

The poles are given by $z^2 + 2z + 4 = 0$

$$\therefore z = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot 4}}{2} = \frac{-2 \pm \sqrt{-12}}{2} = \frac{-2 \pm 2\sqrt{3}i}{2} = -1 \pm \sqrt{3}i$$

Now residue at $z = -1 + i\sqrt{3}$ is

$$\begin{aligned} R_1 &= \lim_{z \rightarrow (-1 + i\sqrt{3})} \frac{z^n [z - (-1 + \sqrt{3}i)]}{[z - (-1 + \sqrt{3}i)] [z - (-1 - \sqrt{3}i)]} \\ &= \frac{[-1 + i\sqrt{3}]^n}{-1 + i\sqrt{3} + 1 + \sqrt{3}i} \\ &= \frac{(-1 + i\sqrt{3})^n}{2\sqrt{3}i} \end{aligned}$$

and residue at $z = -1 - i\sqrt{3}$ is

$$\begin{aligned} R_2 &= \lim_{z \rightarrow (-1 - i\sqrt{3})} \frac{z^n [z - (-1 - i\sqrt{3})]}{[z - (-1 - i\sqrt{3})] [z - (-1 + \sqrt{3}i)]} \\ &= \frac{(-1 - i\sqrt{3})^n}{-1 - i\sqrt{3} + 1 - i\sqrt{3}} \end{aligned}$$

$$= \frac{(-1 - i\sqrt{3})^n}{-2i\sqrt{3}}$$

Now, $f(n) = R_1 + R_2$

$$= \frac{(-1 + i\sqrt{3})^n}{2\sqrt{3}i} + \frac{(-1 - i\sqrt{3})^n}{-2\sqrt{3}i}$$

$$= \frac{1}{2\sqrt{3}i} [(-1 + i\sqrt{3})^n - (-1 - i\sqrt{3})^n]$$

$$= \frac{1}{2\sqrt{3}i} \left[(2)^n \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^n - (2)^n \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)^n \right]$$

$$= \frac{2^n}{2\sqrt{3}i} \left[\left(\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} \right) - \left(\cos \frac{2n\pi}{3} - i \sin \frac{2n\pi}{3} \right) \right]$$

$$= \frac{2^{n-1}}{\sqrt{3}i} \left[\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} - \cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} \right]$$

$$= \frac{2^{n-1}}{i\sqrt{3}} \left[2i \sin \frac{2n\pi}{3} \right]$$

$$= \frac{2^n}{\sqrt{3}} \sin \frac{2n\pi}{3}$$

3. Find Z^{-1} of $F(z) = \frac{z^2 - 3z}{(z-5)(z+2)}$

Let $F(z) = \frac{z^2 - 3z}{(z-5)(z+2)}$, then

$$f(n) = \frac{1}{2\pi i} \int_C z^{n-1} F(z) dz = \frac{1}{2\pi i} \int_C \frac{z^{n-1} (z^2 - 3z)}{(z-5)(z+2)}$$

Clearly, the integrand has poles at $z = 5$ and $z = -2$.

$$\therefore \text{Residue } (R_1)_{z=5} = \lim_{z \rightarrow 5} (z-5) \frac{z^{n-1} (z^2 - 3z)}{(z-5)(z+2)}$$

$$\begin{aligned}
 &= \lim_{z \rightarrow 5} \frac{z^{n+1} - 3z^n}{z + 2} \\
 &= \frac{5^{n+1} - 3 \cdot 5^n}{5 + 2} \\
 &= \frac{5^n \cdot 5 - 3 \cdot 5^n}{7} \\
 &= 5^n \left(\frac{5 - 3}{7} \right) \\
 &= \frac{2 \cdot 5^n}{7} = \frac{2}{7} 5^n
 \end{aligned}$$

$$\begin{aligned}
 \text{And, residue } (R_2)_{z=-2} &= \lim_{z \rightarrow -2} \frac{(z+2)(z^2 - 3z)z^{n-1}}{(z+2)(z-5)} \\
 &= \lim_{z \rightarrow -2} \frac{z^{n+1} - 3z^n}{z-5} \\
 &= \frac{(-2)^{n+1} - 3(-2)^n}{-2-5} \\
 &= \frac{(-2)^{n+1} - 3(-2)^n}{-2-5} \\
 &= \frac{(-2)^n \cdot (-2) - 3(-2)^n}{-7} \\
 &= (-2)^n \left(\frac{-2-3}{-7} \right) = (-2)^n \frac{5}{7}
 \end{aligned}$$

$$\therefore Z^{-1}[F(z)] = f(n) = \frac{5}{7}(-2)^n + \frac{2}{7}5^n$$

4. Find $Z^{-1}\left(\frac{z+1}{z^2-2z+1}\right)$.

Let $F(z) = \frac{z+1}{z^2-2z+1}$, then

$$f(n) = \frac{1}{2\pi i} \int_C z^{n-1} F(z) dz$$

$$= \frac{1}{2\pi i} \int_C z^{n-1} \left(\frac{z+1}{z^2-2z+1} \right) dz$$

$$= \frac{1}{2\pi i} \int_C \frac{z^n + z^{n-1}}{(z-1)^2} dz$$

Clearly, $z=1$ is a pole of order 2.

$$\begin{aligned} \therefore \text{Res}(P)_{z=1} &= \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \frac{z^n + z^{n-1}}{(z-1)^2} \right] \\ &= \lim_{z \rightarrow 1} \frac{d}{dz} (z^n + z^{n-1}) \\ &= \lim_{z \rightarrow 1} [nz^{n-1} + (n-1)z^{n-2}] \\ &= n + n - 1 \\ &= 2n - 1 \\ \therefore Z^{-1}\{f(z)\} &= f(n) = 2n - 1 \end{aligned}$$

6. Find $Z^{-1} \left[\frac{2z^2 + z}{(z-2)^2(z-1)} \right]$

Let $f(z) = \frac{2z^2 + z}{(z-2)^2(z-1)}$, then

$$f(n) = \frac{1}{2\pi i} \int_C \frac{z^{n-1} (2z^2 + z)}{(z-2)^2(z-1)} dz$$

Clearly, the poles are $z=1$ of order 1 and $z=2$ of order 2.

$$\begin{aligned} \text{Now, residue } (P)_{z=1} &= \lim_{z \rightarrow 1} (z-1) \frac{(2z^2 + z) z^{n-1}}{(z-2)^2(z-1)} \\ &= \lim_{z \rightarrow 1} \frac{2z^{n+1} + z^n}{(z-2)^2} \\ &= \frac{2+1}{(1-2)^2} = 3 \end{aligned}$$

$$\begin{aligned} \text{and residue } (P)_{z=2} &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[(z-2)^2 \frac{(2z^2 + z) z^{n-1}}{(z-2)^2(z-1)} \right] \\ &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{(2z^2 + z) z^{n-1}}{z-1} \right] \end{aligned}$$

$$\begin{aligned}
 &= \lim_{z \rightarrow 2} \left[\frac{(z-1)(2(n+2)z^{n+1} + nz^{n-1}) - (2z^{n+2} + z^n) \cdot 1}{(z-1)^2} \right] \\
 &= \lim_{z \rightarrow 2} \left[\frac{(z-1)(2(n+2)z^{n+1} + nz^{n-1}) - 2z^{n+2} - z^n}{(z-1)^2} \right] \\
 &= \left[\frac{2(n+2)2^{n+1} + n2^{n-1} - 2 \cdot 2^{n+2} - 2^n}{1} \right] \\
 &= 2(n+2)2^n \cdot 2 + n2^{n-1} - 2 \cdot 2^n \cdot 2^2 - 2^n \\
 &= 4(n+2)2^n - 8 \cdot 2^n - 2^n + n2^{n-1} \\
 &= 2^n(4n + 8 - 8 - 1) + n2^{n-1} \\
 &= 2^n(4n - 1) + n2^{n-1} \\
 &= 2^n(4n - 1) + \frac{n}{2} \cdot 2^n \\
 &= 2^n \left(4n - 1 + \frac{n}{2} \right) \\
 &= 2^n \left(\frac{8n - 2 + n}{2} \right) \\
 &= 2^n \left(\frac{9n - 2}{2} \right)
 \end{aligned}$$

$$\therefore f(n) = \text{sum of the residues} = 3 + \frac{9n - 2}{2} 2^n$$

6. Find Z inverse of $\frac{z}{(z-1)(z^2+1)}$. *Ans*

Let $F(z) = \frac{z}{(z-1)(z^2+1)}$, then

$$f(n) = \frac{1}{2\pi i} \oint_C z^{n-1} f(z) dz$$

$$= \frac{1}{2\pi i} \oint_C \frac{z^n}{(z-1)(z^2+1)} dz$$

$$= \frac{1}{2\pi i} \int_C \frac{z^n}{(z-1)(z-i)(z+i)} dz$$

Clearly, the poles are $z = 1$, $z = i$ and $z = -i$.

Residue at $z = 1$ is

$$(R_1)_{z=1} = \lim_{z \rightarrow 1} (z-1) \left[\frac{z^n}{(z-1)(z-i)(z+i)} \right] = \frac{(1)^n}{2} = \frac{1}{2}$$

Again, Residue at $z = i$ is

$$(R_2)_{z=i} = \lim_{z \rightarrow i} \frac{(z-i) z^n}{(z-1)(z+i)(z-i)} = \frac{(i)^n}{(i-1)2i}$$

and also residue at $z = -i$ is

$$(R_3)_{z=-i} = \lim_{z \rightarrow -i} \left[\frac{(z+i) z^n}{(z-1)(z+i)(z-i)} \right] = \frac{(-i)^n}{(-i-1)(-2i)}$$

Now $Z^{-1} F(z) = f(n) = \text{sum of residues}$

$$= R_1 + R_2 + R_3$$

$$= \frac{1}{2} + \frac{(i)^n}{(i-1)2i} + \frac{(-i)^n}{(-i-1)(-2i)}$$

$$= \frac{1}{2} + 2 \operatorname{Re} \left[\frac{i^n}{(i-1)2i} \right]$$

$$= \frac{1}{2} - \operatorname{Re} \left[\frac{i^n}{(i+1)} \right]$$

$$= \frac{1}{2} - \frac{1}{2} \operatorname{Re} [(1-i) i^n]$$

$$= \frac{1}{2} - \frac{1}{2} \operatorname{Re} \left[(1-i) \left(\cos \frac{n\pi}{2} + i \sin \frac{n\pi}{2} \right) \right]$$

$$= \frac{1}{2} - \frac{1}{2} \left(\cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right)$$

$$= \frac{1}{2} \left(1 - \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right)$$

Find the inverse of Z-transform by using convolution property.

(i) $\frac{z^2}{(z-2)(z-3)}$

(ii) $\frac{z^2}{(z-a)^2}$

$$\begin{aligned} Z^{-1} \left[\frac{z^2}{(z-2)(z-3)} \right] &= Z^{-1} \left(\frac{z}{z-2} \right) * Z^{-1} \left(\frac{z}{z-3} \right) \\ &= 2^n * 3^n \\ &= \sum_{k=0}^n 2^{n-k} 3^k \\ &= 2^n \sum_{k=0}^n 2^{-k} 3^k \\ &= 2^n \sum_{k=0}^n \left(\frac{3}{2} \right)^k \\ &= 2^n \left[1 + \frac{3}{2} + \left(\frac{3}{2} \right)^2 + \dots + \left(\frac{3}{2} \right)^n \right] \\ &= 2^n \left[\frac{\left(\frac{3}{2} \right)^{n+1} - 1}{\frac{3}{2} - 1} \right] \\ &= 2^{n+1} \left[\frac{3^{n+1}}{2^{n+1}} - 1 \right] \\ &= 3^{n+1} - 2^{n+1} \end{aligned}$$

ii. $Z^{-1} \left[\frac{z^2}{(z-a)^2} \right] = Z^{-1} \left(\frac{z}{z-a} \right) * Z^{-1} \left(\frac{z}{z-a} \right)$

$$\begin{aligned} &= a^n * a^n \\ &= \sum_{k=0}^n a^{n-k} a^k \\ &= \sum_{k=0}^n (a^{-k} a^k) a^n \end{aligned}$$

$$= d + d + \dots + d$$

$$= (n + 1)d$$

7.14 Difference Equation

The difference equation is the equation between the difference of an unknown function. It describes the pattern of some periodic function with reference to two or more consecutive periods.

The solution of difference equation is any function which satisfies the given equation. The general solution of a difference equation is defined as the solution which involves as many arbitrary constants as the order of the difference equation. The particular solution is a solution obtained from the general solution by assigning particular values to periodic constants.

Here, we use Z-transform to solve the difference equation. Now we state and prove the following theorem.

Theorem. If $Z(y_p) = \bar{y}$, then $Z(y_{p+k}) = z \left[\bar{y} - y_0 - \frac{z_1}{z} - \dots - \frac{z_{k-1}}{z^{k-1}} \right]$

Proof.

By definition, $Z(y_{p+k}) = \sum_{p=0}^{\infty} y_{p+k} z^{-p}$

$$= \sum_{p=0}^{\infty} y_{p+k} z^{-p-k} z^k$$

$$= z^k \sum_{p=0}^{\infty} y_{p+k} z^{-p-k}$$

Let $p = n + k$ then as $n = 0$, $p = k$ and when $n = \infty$, $p = \infty$

$$Z(y_{p+k}) = z^k \sum_{p=k}^{\infty} y_p z^{-p}$$

$$= z^k \left[\sum_{p=0}^{\infty} y_p z^{-p} - \sum_{p=0}^{k-1} y_p z^{-p} \right]$$

$$= z^k \left[\bar{y} - y_0 - \frac{z_1}{z} - \frac{z_2}{z^2} - \dots - \frac{z_{k-1}}{z^{k-1}} \right], \text{ where } \bar{y} = Z(y_p)$$

Special Cases.

1. If $k = 1$, then

$$Z(y_{n+1}) = z(\bar{y} - y_0)$$

2. If $k = 2$, then

$$Z(y_{n+2}) = z^2\left(\bar{y} - y_0 - \frac{y_1}{z}\right)$$

3. If $k = 3$, then

$$Z(y_{n+3}) = z^3\left(\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2}\right)$$

Examples

1. Solve the difference equation $y_{n+2} + 4y_{n+1} + 3y_n = 0$ given that $y_0 = 1, y_1 = 0$.

We have $y_{n+2} + 4y_{n+1} + 3y_n = 0$

Taking Z-transform on both sides,

$$Z(y_{n+2}) + 4Z(y_{n+1}) + 3Z(y_n) = 0$$

$$\text{or, } z^2\left(\bar{y} - y_0 - \frac{y_1}{z}\right) + 4z(\bar{y} - y_0) + 3\bar{y} = 0$$

$$\text{or, } z^2(\bar{y} - 1) + 4z(\bar{y} - 1) + 3\bar{y} = 0$$

$$\text{or, } \bar{y}(z^2 + 4z + 3) = z^2 + 4z$$

$$\therefore \bar{y} = \frac{z^2 + 4z}{z^2 + 4z + 3}$$

$$\text{or, } Z(y_n) = \frac{z^2 + 4z}{z^2 + 4z + 3}$$

$$\therefore y_n = Z^{-1}\left(\frac{z^2 + 4z}{z^2 + 4z + 3}\right) \quad (1)$$

$$\text{Let } F(z) = \frac{z^2 + 4z}{z^2 + 4z + 3} = \frac{z^2 + 4z}{(z+1)(z+3)}, \text{ and}$$

$$\text{Let } \frac{z^2 + 4z}{(z+1)(z+3)} = \frac{Az}{(z+1)} + \frac{Bz}{(z+3)}$$

$$z^2 + 4z = Az(z + 3) + Bz(z + 1)$$

When $z = -1$, $A = \frac{3}{2}$ and when $z = -3$, $B = -\frac{1}{2}$

$$\therefore \frac{z^2 + 4z}{(z + 1)(z + 3)} = \frac{3}{2} \left(\frac{z}{z + 1} \right) - \frac{1}{2} \left(\frac{z}{z + 3} \right)$$

From (1), $y_n = \frac{3}{2} Z^{-1} \left(\frac{z}{z + 1} \right) - \frac{1}{2} Z^{-1} \left(\frac{z}{z + 3} \right) = \frac{3}{2} (-1)^n - \frac{1}{2} (-3)^n$

2. Solve $y_{n+2} - 5y_{n+1} + 6y_n = 4^n$, $y_0 = y_1 = 0$.

We have $y_{n+2} - 5y_{n+1} + 6y_n = 4^n$

Taking Z-transform on both sides

$$Z(y_{n+2}) - 5Z(y_{n+1}) + 6Z(y_n) = Z(4^n)$$

or, $z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 5z(\bar{y} - y_0) + 6\bar{y} = \frac{z}{z - 4}$

or, $z^2 (\bar{y}) - 5z(\bar{y}) + 6\bar{y} = \frac{z}{z - 4}$

or, $\bar{y} (z^2 - 5z + 6) = \frac{z}{z - 4}$

or, $\bar{y} = \frac{z}{(z - 4)(z - 2)(z - 3)}$

or, $Z(y_n) = \frac{z}{(z - 4)(z - 2)(z - 3)}$

$\therefore y_n = Z^{-1} \left[\frac{z}{(z - 4)(z - 2)(z - 3)} \right]$

$$= \frac{1}{2\pi i} \int_c \frac{z^n dz}{(z - 4)(z - 2)(z - 3)}$$

The poles are $z = 2, 3$ and 4

Now, residue (R_1) at $z = 4$ is

$$(R_1)_{z=4} = \lim_{z \rightarrow 4} (z-4) \frac{z^n}{(z-4)(z-2)(z-3)} = \frac{1}{2} (4)^n$$

Residue (R_2) at $z = 3$ is

$$(R_2)_{z=3} = \lim_{z \rightarrow 3} (z-3) \frac{z^n}{(z-4)(z-2)(z-3)} = -3^n$$

Residue (R_3) at $z = 2$ is

$$(R_3)_{z=2} = \lim_{z \rightarrow 2} (z-2) \frac{z^n}{(z-4)(z-2)(z-3)} = \frac{1}{2} (2)^n$$

$$\therefore y_n = \text{sum of residues} = \frac{1}{2} (4)^n - 3^n + \frac{1}{2} (2)^n$$

$$3. \text{ Solve } y_{n+2} + 2y_{n+1} + y_n = n \quad y_0 = y_1 = 0.$$

The given equation is

$$y_{n+2} + 2y_{n+1} + y_n = n$$

Taking Z-transform on both sides

$$Z(y_{n+2}) + 2Z(y_{n+1}) + Z(y_n) = Z(n)$$

$$\text{or, } z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) + 2z(\bar{y} - y_0) + (\bar{y}) = \frac{z}{(z-1)^2}$$

$$\text{or, } z^2(\bar{y}) + 2z(\bar{y}) + (\bar{y}) = \frac{z}{(z-1)^2}$$

$$\text{or, } (z^2 + 2z + 1) \bar{y} = \frac{z}{(z-1)^2}$$

$$\text{or, } \bar{y} = \frac{z}{(z-1)^2 (z^2 + 2z + 1)}$$

$$= \frac{z}{(z-1)^2 (z+1)^2}$$

$$\therefore y_n = Z^{-1} \left[\frac{z}{(z-1)^2 (z+1)^2} \right] = \frac{1}{2\pi i} \int_C \frac{z^e dz}{(z-1)^2 (z+1)^2}$$

Here the integrand has poles $z = 1$ and $z = -1$ both of order 2.

Also $f(n) = \text{sum of residues} = R_1 + R_2$

$$\begin{aligned}\text{Now, residue } (R_1)_{z=1} &= \lim_{z \rightarrow 1} \left[\frac{d}{dz} (z-1)^2 \frac{z^n}{(z-1)^2 (z+1)^2} \right] \\ &= \lim_{z \rightarrow 1} \left[\frac{d}{dz} \frac{z^n}{(z+1)^2} \right] \\ &= \lim_{z \rightarrow 1} \frac{(z+1)^2 n z^{n-1} - z^n \cdot 2(z+1)}{[(z+1)^2]^2} \\ &= \frac{4n-4}{16} \\ &= \frac{n-1}{4}\end{aligned}$$

$$\begin{aligned}\text{Again, residue } (R_2)_{z=-1} &= \lim_{z \rightarrow -1} \left[\frac{d}{dz} (z+1)^2 \frac{z^n}{(z-1)^2 (z+1)^2} \right] \\ &= \lim_{z \rightarrow -1} \left[\frac{d}{dz} \frac{z^n}{(z-1)^2} \right] \\ &= \lim_{z \rightarrow -1} \left[\frac{(z-1)^2 n z^{n-1} - z^n \cdot 2(z-1)}{[(z-1)^2]^2} \right] \\ &= \frac{4n(-1)^{n-1} + 4(-1)^n}{16} \\ &= \frac{-4n(-1)^n + 4(-1)^n}{16} \\ &= \frac{(-1)^n (4-4n)}{16} \\ &= \frac{(-1)^n (1-n)}{4}\end{aligned}$$

$$\therefore y_n = R_1 + R_2$$

$$= \frac{n-1}{4} + \frac{(-1)^n (1-n)}{4}$$

$$= \frac{n-1}{4} [1 - (-1)^n]$$

4. Solve $U_{x+2} - 2\cos\alpha U_{x+1} + U_x = 0$ $U_0 = 1, U_1 = \cos\alpha$

We have $U_{x+2} - 2\cos\alpha U_{x+1} + U_x = 0$

Taking Z-transform on both sides

$$Z(U_{x+2}) - 2\cos\alpha Z(U_{x+1}) + Z(U_x) = 0$$

$$\text{or, } z^2 \left(\bar{U} - U_0 - \frac{U_1}{z} \right) + 2\cos\alpha z (\bar{U} - U_0) + \bar{U} = 0$$

$$\text{or, } z^2 \left(\bar{U} - 1 - \frac{\cos\alpha}{z} \right) + 2\cos\alpha z (\bar{U} - 1) + \bar{U} = 0$$

$$\text{or, } \bar{U} (z^2 - 2\cos\alpha z + 1) - z^2 + z\cos\alpha = 0$$

$$\text{or, } \bar{U} = \frac{z(z - \cos\alpha)}{z^2 - 2\cos\alpha z + 1}$$

$$\text{Now, } U = Z^{-1} \left[\frac{z(z - \cos\alpha)}{z^2 - 2z\cos\alpha + 1} \right] = \frac{1}{2\pi i} \int \frac{z^2 (z - \cos\alpha)}{z^2 - 2z\cos\alpha + 1} dz$$

$$= \frac{1}{2\pi i} \int \frac{z^{x+1} - z^x \cos\alpha}{(z - e^{i\alpha})(z - e^{-i\alpha})} dz$$

Clearly, $e^{i\alpha}$ and $e^{-i\alpha}$ are simple poles.

Now, residue at $e^{i\alpha}$ is

$$\begin{aligned} R_1 &= \lim_{z \rightarrow e^{i\alpha}} \left(\frac{z^{x+1} - z^x \cos\alpha}{z - e^{-i\alpha}} \right) \\ &= \frac{e^{i(x+1)\alpha} - e^{ix\alpha} \cos\alpha}{e^{i\alpha} - e^{-i\alpha}} = \frac{e^{i(x+1)\alpha} - e^{ix\alpha} \cos\alpha}{2i \sin\alpha} \end{aligned}$$

Similarly, residue at $z = e^{-i\alpha}$ is

$$R_2 = \lim_{z \rightarrow e^{-i\alpha}} \frac{e^{-i(x+1)\alpha} - e^{-ix\alpha} \cos\alpha}{-2i \sin\alpha}$$

$$\begin{aligned} \therefore U_x &= R_1 + R_2 = \frac{e^{i(x+1)\alpha} - e^{ix\alpha} \cos\alpha}{2i \sin\alpha} + \frac{e^{-i(x+1)\alpha} - e^{-ix\alpha} \cos\alpha}{-2i \sin\alpha} \\ &= 2\text{Re} \left[\frac{e^{i(x+1)\alpha} - e^{ix\alpha} \cos\alpha}{2i \sin\alpha} \right] \end{aligned}$$

$$= -\frac{1}{\sin \alpha} \operatorname{Re} [i \{e^{i(n+1)\alpha} - e^{i n \alpha} \cos \alpha\}]$$

$$= \cos n \alpha$$

5. Solve $y_{n+2} - 7y_{n+1} + 12y_n = 2n$, $y_0 = 0$, $y_1 = 0$

We have $y_{n+2} - 7y_{n+1} + 12y_n = 2n$

Taking Z-transform on both sides

$$Z(y_{n+2}) - 7Z(y_{n+1}) + 12Z(y_n) = Z(2n)$$

$$\text{or, } z^2(\bar{y} - y_0 - \frac{y_1}{z}) - 7z(\bar{y} - y_0) + 12\bar{y} = \frac{2z}{(z-1)^2}$$

$$\text{or, } z^2(\bar{y}) - 7z(\bar{y}) + 12\bar{y} = \frac{2z}{(z-1)^2}$$

$$\text{or, } \bar{y}(z^2 - 7z + 12) = \frac{2z}{(z-1)^2}$$

$$\Rightarrow \bar{y} = \frac{2z}{(z-1)^2(z^2 - 7z + 12)}$$

$$\text{or, } \bar{y} = \frac{2z}{(z-1)^2(z-3)(z-4)}$$

$$\text{Now, } y_n = Z^{-1} \left[\frac{2z}{(z-1)^2(z-3)(z-4)} \right]$$

$$= \frac{1}{2\pi i} \int_C \frac{2z^n dz}{(z-1)^2(z-3)(z-4)} = \text{sum of residues}$$

The poles are $z = 1$ (order 2), $z = 3$ and $z = 4$.

Now the residue at $z = 1$ is

$$(R_1)_{z=1} = \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left\{ \frac{2z^n (z-1)^2}{(z-1)^2(z-3)(z-4)} \right\} \right]$$

$$= \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{2z^n}{z^2 - 7z + 12} \right) \right]$$

$$= \lim_{z \rightarrow 1} \left[\frac{(z^2 - 7z + 12) 2nz^{n-1} - 2z^n (2z - 7)}{(z^2 - 7z + 12)^2} \right]$$

$$= \frac{12n + 10}{36} = \frac{n}{3} + \frac{5}{18}$$

Residue at $z = 3$ is

$$(R_2)_{z=3} = \lim_{z \rightarrow 3} \frac{(z-3) 2z^n}{(z-1)^2 (z-3) (z-4)} = \frac{2(3)^n}{4(-1)} = -\frac{1}{2} (3)^n$$

Similarly, Residue at $z = 4$ is

$$(R_3)_{z=4} = \lim_{z \rightarrow 4} \frac{(z-4) 2z^n}{(z-1)^2 (z-3) (z-4)} = \frac{2(4)^n}{9}$$

$$\therefore y_n = R_1 + R_2 + R_3$$

$$= \frac{5}{18} + \frac{n}{3} - \frac{1}{2} 3^n + \frac{2}{9} (4)^n$$

6. Solve $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$ $y_0 = 0, y_1 = 1$

We have $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$

Taking Z-transform on both sides

$$Z(y_{n+2}) - 4Z(y_{n+1}) + 4Z(y_n) = Z(2^n)$$

$$\text{or, } z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 4z(\bar{y} - y_0) + 4\bar{y} = \frac{z}{z-2}$$

$$\text{or, } z^2 \left(\bar{y} - \frac{1}{z} \right) - 4z(\bar{y}) + 4\bar{y} = \frac{z}{z-2}$$

$$\text{or, } \bar{y}(z^2 - 4z + 4) - z = \frac{z}{z-2}$$

$$\text{or, } \bar{y}(z^2 - 4z + 4) = \frac{z}{z-2} + z$$

$$\text{or, } \bar{y} = \frac{z + z(z-2)}{(z-2)(z^2 - 4z + 4)}$$

$$\text{or, } \bar{y} = \frac{z^2 - z}{(z-2)^3}$$

$$\text{or, } Z(y_n) = \frac{z^2 - z}{(z-2)^3}$$

$$\Rightarrow y_n = Z^{-1} \left[\frac{z^2 - z}{(z-2)^3} \right]$$

$$= \frac{1}{2\pi i} \int \frac{z^{n-1} (z^2 - z) dz}{(z-2)^3}$$

$$= \frac{1}{2\pi i} \int \frac{(z^{n+1} - z^n) dz}{(z-2)^3} = \text{sum of residues}$$

Clearly, $z = 2$ is a pole of order 3.

∴ Residue at $z = 2$ is

$$\begin{aligned} (R)_{z=2} &= \lim_{z \rightarrow 2} \left[\frac{1}{2!} \frac{d^2}{dz^2} \left\{ (z-2)^3 \frac{z^{n+1} - z^n}{(z-2)^3} \right\} \right] \\ &= \lim_{z \rightarrow 2} \left[\frac{1}{2} \frac{d^2}{dz^2} (z^{n+1} - z^n) \right] \\ &= \lim_{z \rightarrow 2} \frac{1}{2} \frac{d}{dz} [(n+1)z^n - nz^{n-1}] \\ &= \lim_{z \rightarrow 2} \frac{1}{2} [(n+1)n z^{n-1} - n(n-1)z^{n-2}] \\ &= \frac{1}{2} [(n+1)n 2^{n-1} - n(n-1)2^{n-2}] \\ &= \frac{1}{2} [(n+1)n 2^{n-2} \cdot 2 - n(n-1)2^{n-2}] \\ &= \frac{1}{2} [(2n(n+1) - n(n-1)) 2^{n-2}] \\ &= \frac{1}{2} [(2n^2 + 2n - n^2 + n) 2^{n-2}] \\ &= \frac{1}{2} [(n^2 + 3n) 2^{n-2}] = 2^{n-3} n(n+3) \\ \therefore y_1 &= n(n+3) 2^{n-3} \end{aligned}$$

7. Solve $U_{n+2} + U_n = 5 \cdot 2^n$, $U_0 = 1, U_1 = 0$

We have $U_{n+2} + U_n = 5 \cdot 2^n$

Taking Z-transform on both sides

$$Z(U_{n+2}) + ZU_n = 5 Z(2^n)$$

$$\text{or, } z^2 \left(U - U_0 - \frac{U_1}{z} \right) + U = 5 \left(\frac{z}{z-2} \right)$$

$$\text{or, } (z^2 + 1)U - z^2 = \frac{5z}{z-2}$$

$$\text{or, } (z^2 + 1)U = \frac{5z}{z-2} + z^2 = \frac{5z + z^3 - 2z^3}{z-2}$$

$$\text{or, } \bar{U} = \frac{z^3 - 2z^2 + 5z}{(z-2)(z^2+1)}$$

$$\therefore U_x = Z^{-1} \left[\frac{z^3 - 2z^2 + 5z}{(z-2)(z^2+1)} \right]$$

$$\text{Let } F(z) = \frac{z^3 - 2z^2 + 5z}{(z-2)(z^2+1)} \quad \text{and let}$$

$$\frac{z^3 - 2z^2 + 5z}{(z-2)(z^2+1)} = \frac{Az}{z-2} + \frac{Bz^2 + Cz}{z^2+1}$$

$$\text{or, } z^3 - 2z^2 + 5z = Az(z^2+1) + (Bz^2 + Cz)(z-2)$$

When $z = 2$, $A = 1$.

Comparing the coefficients of z and z^2 , we have

$$A - 2C = 5 \quad \therefore C = -2$$

$$-2B + C = -2 \quad \therefore B = 0$$

$$\therefore F(z) = \frac{z}{z-2} + \frac{-2z}{z^2+1}$$

$$\therefore U_x = Z^{-1} \left[\left(\frac{z}{z-2} \right) - \left(\frac{2z}{z^2+1} \right) \right]$$

$$= 2^x - 2 \sin \frac{x\pi}{2}$$

Exercise 7.3

1. Find the Z-transform of the followings.

a. n

b. $\frac{1 - e^{-bt}}{b}$

c. t^2

d. $\cos 4n\pi$

e. $e^{-at} - e^{-bt}$

f. $\sin w_0 u(n)$

g. $(1+n) u_n$

h. $[4(3)^n - 2^n] u_n$

i. $t^2 e^{-t}$

j. $e^{2t} \cos t$

k. $\delta(n-3)$

2. Find the inverse Z-transform of the followings.

a. $\frac{z}{(z-2)(z-3)}$

b. $\frac{z}{z^2 - 3z + 2}$

c. $\frac{10z + 5}{(z-1)\left(z - \frac{1}{5}\right)}$

d. $\frac{z^2 - 3z}{(z-5)(z+2)}$

e. $\frac{6z^3}{(2z-1)(3z+2)}$

g. $\frac{z}{(z-1)^2(z+1)}$

i. $\frac{2z^3+z}{(z-2)^2(z-1)}$

k. $\frac{z^2-4z}{(z-2)^2}$

m. $\frac{z}{z^2-2z+2}$

o. $\frac{z(z+1)}{(z-1)^3}$

q. $\frac{9z^3}{(3z-1)^2(z-2)}$

s. $\frac{z+2}{z^2-5z+6}$

u. $\frac{z}{z^2-2z+2}$

f. $\frac{8z^2}{(2z-1)(4z-1)}$

h. $\frac{z(z+2)}{(z-1)^2}$

j. $\frac{z(z^2-1)}{(z^2+1)^2}$

l. $\frac{3z^2+2z+1}{z^2+3z+2}$

n. $\frac{z^2-3z}{z^2-3z-10}$

p. $\frac{z^2+z+2}{(z-1)(z^2-z+1)}$

r. $\frac{z^3-20z}{(z-4)(z-2)^3}$

t. $\frac{z^2-3}{(z+2)(z^2+1)}$

3. Find the inverse Z-transform of the followings.

a. $\frac{z^{-1}(1-z^{-2})}{(1+z^{-2})^2}$

b. $\frac{1+z^{-1}}{1-z^{-1}+\frac{1}{2}z^{-2}}$

c. $\frac{1}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}$

d. $\frac{1+6z^{-1}+z^{-3}}{(1+z^{-1})\left(1+\frac{1}{5}z^{-1}\right)}$

4. Solve the following difference equations.

a. $y_{k+1} + y_k = 1,$

$y_0 = 0$

b. $y_{n+2} - 4y_{n+1} + 4y_n = 0,$

$y_0 = 1, y_1 = 0$

c. $y_{n+2} + y_n = 1,$

$y_0 = y_1 = 0$

d. $y_{x+2} - 3y_{x+1} + 2y_x = 0,$

$y_0 = 0, y_1 = 1$

e. $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$

$y_0 = y_1 = 0$

f. $y_{n+2} - 5y_{n+1} + 6y_n = 36,$

$y_0 = y_1 = 0$

g. $x_{n+1} - y_n = 1, y_{n+1} + x_n = 0,$

$x_0 = 0, y_0 = -1$

h. $y_{n+2} - 3y_{n+1} + 2y_n = 4^n,$

$y_0 = 0, y_1 = 1$

i. $y_{n+2} - 2\cos\alpha y_{n+1} + y_n = 0, \quad y_0 = 1, y_1 = \cos\alpha$
 j. $y_{k+2} + 2y_{k+1} + y_k = k, \quad y_0 = 0, y_1 = 0$

Solve the difference equation $y_{n+2} = y_{n+1} + y_n$ where

$y_0 = 0, y_1 = 1$ such that $\frac{y_{n+1}}{y_n} = \frac{1 + \sqrt{5}}{2}$ where $n \rightarrow \infty$.

Answers

a. $\frac{z}{(z-1)^2}$

c. $\frac{T^2 z(z+1)}{(z-1)^3}$

e. $\frac{(e^{-aT} - e^{-bT})z}{(z - e^{-aT})(z - e^{-bT})}$

g. $\frac{z^2}{(z-1)^2}$

i. $\frac{T^2 (ze^T)(ze^T + 1)}{(ze^T - 1)^3}$

k. $\frac{1}{z^3}$

b. $\frac{1}{b} \left[\frac{z}{z-b} - \frac{z}{z-e^{-bT}} \right]$

d. $\frac{z(z - \cos 4\pi)}{z^2 - 2z \cos 4\pi + 1}$

f. $\frac{z \sin w_0}{z^2 - 2z \cos w_0 + 1}$

h. $\frac{3z^2 - 5z}{(z-3)(z-2)}$

j. $\frac{(ze^{-2T})(ze^{-2T} - \cos T)}{z^2 e^{-4T} - 2ze^{-2T} \cos T + 1}$

a. $3^n - 2^n$

c. $\frac{1}{4} \left[75 - \frac{7}{5^{n-2}} \right]$

e. $\frac{3}{7} \left(\frac{1}{2} \right)^n + \frac{4}{7} \left(\frac{-2}{3} \right)^n$

g. $\frac{1}{4} (-1)^n - \frac{1}{4} + \frac{1}{2} n$

i. $9n(2)^{n-1} - (2)^n + 3$

k. $(2)^n - 2n(2)^{n-1}$

m. $(\sqrt{2})^2 \sin \frac{n\pi}{4}$

b. $1 - 2^n$

d. $\frac{2}{7} (5)^n + \left(\frac{5}{7} \right) (-2)^n$

f. $2 \left[\left(\frac{1}{2} \right)^n - \left(\frac{1}{4} \right)^n \right]$

h. $3n + 1$

j. $\frac{n}{2} [(i)^{n-1} + (-i)^{n-1}]$

l. $2(-1)^{n-1} - 9(-2)^{n-1}$

n. $\frac{1}{7} [2(5)^n + 5(-2)^n]$

o. n^2

p. $4 - 2\sqrt{3} \sin \frac{n\pi}{\sqrt{3}} + \frac{4}{\sqrt{3}} \sin \frac{(n-1)\pi}{3}$

q. $\frac{9}{25} (2)^{n+2} - \frac{1}{5} (n+2) \frac{1}{3^n}$

r. $2^{n-1} + n^2 2^n - 4^n$

s. $\frac{5}{3} (3^n) - 2^{n+1}$

3. a. $(-1)^n 2n + 1$

b. $\left(\frac{1}{2}\right)^n \left(\cos \frac{n\pi}{4} + 3 \sin \frac{n\pi}{4}\right)$

c. $2 - \left(\frac{1}{2}\right)^n$

d. $\frac{3}{4} (-1)^n - \frac{13}{2} (-1)^n \left(\frac{1}{5}\right)^n$

4. a. $y_k = \frac{1}{2} - \frac{1}{2} (-1)^k$

b. $y_n = 2^n (1 - 2n)$

c. $y_n = \frac{1}{2} \left[1 - \left(\cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right) \right]$

d. $y_n = 2^x - 1$

e. $y_n = \frac{1}{25} (2)^n - \frac{1}{25} (-3)^n - \frac{1}{5} n (-3)^n$

f. $y_n = 18(3)^n - 36(2)^n + 18$

g. $y_n = \frac{1}{2} \left[\sin \frac{n\pi}{2} - \cos \frac{n\pi}{2} - 1 \right], \quad x_n = \frac{1}{2} \left[1 - \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right]$

h. $y_n = -\frac{2}{3} + \frac{1}{3} (2)^{2n-1} - 2^{n-1}$

i. $y_l = \cos a$

j. $y_k = \frac{1}{4} [k - 1 + (-1)^k + k(-1)^k]$

5. $y_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$